

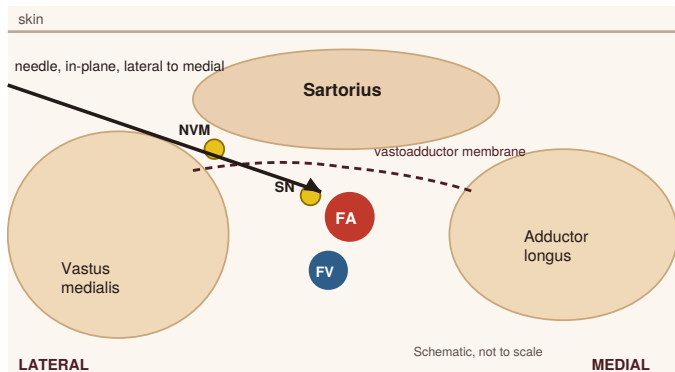
Adductor canal block and catheters

Station 4 notes · Lower limb anterior · ARACUS 2026
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1. Anatomy you must see



- **Probe:** linear, transverse, anteromedial thigh, roughly midway between inguinal crease and patella.
- **Sonoanatomy:** round femoral artery (FA) under sartorius; vein (FV) deep and often compressed; vastus medialis lateral; adductor longus deep and medial. The saphenous nerve (SN) is the bright dot immediately lateral to the artery.
- **Apex of the femoral triangle:** where the medial borders of sartorius and adductor longus meet. Proximal to it is femoral triangle; distal to it is the true adductor canal. In practice local anaesthetic a few cm either side gives a similar block.

Two nerves, two tunnels. The knee's anteromedial capsule is largely supplied by the **saphenous nerve** and the **nerve to vastus medialis (NVM)**. The SN lies beside the artery deep to the **vastoadductor membrane (VAM)**. The NVM lies in the plane between sartorius and vastus medialis, outside that membrane. A saphenous-only block numbs the medial calf but gives limited knee analgesia.

Level matters

Too proximal	Too distal
Adductor longus bulky, sartorius slides laterally off the artery. More femoral motor fibres reached: quadriceps weakness.	Adductor longus gone, artery drops away from sartorius towards the hiatus. SN and NVM diverge: less reliable block ; distal spread can reach the popliteal fossa.

2. Single-shot technique

Needle in-plane from lateral. Enter the skin **several cm lateral to the probe** for a shallow, straight path through both targets.

Two-compartment approach	Periarterial approach
Nerve stimulator on. A local sartorius twitch, then a true vastus medialis twitch (hand on distal medial thigh) marks the NVM. Inject 10 mL there. Advance, feel resistance then a pop through the VAM , inject 10 mL beside the artery for the SN. Total 20 mL.	Aim deep to the VAM at the 12 o'clock aspect of the artery. Relies on the volume spreading back to the NVM. A more distal level spares the quadriceps further.

You are in the right compartment when: spread outlines the arterial wall; the artery squashes as you inject; you felt a tactile pop. **Not yet:** a linear sheet of fluid under sartorius, still superficial to the VAM.

Protect the NVM. In a cadaver study, more than half of needles aimed blindly at the SN touched or pierced the NVM, and isolated vastus medialis wasting after this block has been reported. Nerve stimulation helps you find it and avoid it. **Saphenous only** (e.g. with a sciatic block for ankle surgery): come just lateral to the artery and deliberately avoid the NVM.

3. Catheter technique

Oblique approach as taught by Ki-Jinn Chin, placed before total knee replacement.

1. **Position:** supine, leg slightly abducted and externally rotated if comfortable. No knee flexion or exaggerated frog leg needed.
2. **Entry:** one to two hand breadths above the patella so the dressing stays clear of the surgical drapes. Short subcutaneous tunnel before the canal.
3. **Scout scan**, then rotate the probe (lateral end towards the head) for an oblique view: artery and sartorius elongate. Find the lateral corner of sartorius where it meets vastus medialis.
4. **Skin local** 2 to 3 cm from the probe edge, on the planned track.
5. **Needle:** steep through skin, then flatten. Enter at the lateral edge of sartorius and run in the **sartorius / vastus medialis fascial plane**. **Not through the muscle:** sartorius contraction on knee flexion pulls catheters out.
6. **Cross the VAM well lateral to the artery**, then flatten and advance to the artery's 12 o'clock. Hydrodissect ahead of the tip; tilt the curved Tuohy bevel down to depress the artery. If the tip keeps diving, **ease off probe pressure**.
7. **Loading bolus** of long-acting local anaesthetic opens the space. Scan dynamically; identify and avoid the SN.
8. **Lift the probe gently** (tissue rebound drags needles). Thread the catheter just past the needle tip until resistance; **never force it**.
9. **Image the tip before removing the needle.** You cannot advance afterwards. Jiggling the catheter shows the tip.
10. **Withdraw the needle** with gentle forward pressure on the catheter. Test inject: some spread along the track is normal; you want **periarterial** spread.

Device notes

Set	What to know
Arrow StimuCath catheter through Tuohy	Tip leaves the needle at the 10 cm mark. Push the catheter fully through the connector's internal diaphragm before locking, or nothing will inject.
Pajunk SonoLong 100 mm Tuohy	Blunt needle: use the supplied cutting stylet for a skin nick (can ooze and spoil the dressing) or enter near-perpendicular. Tip leaves at 13 cm ; thread only 1 to 2 cm more. Dry the catheter end before fitting the hub.
Pajunk E-Cath / E-Cath PLUS catheter over needle	Bolus through the stylet. The standard catheter tip only reaches where the stylet tip was; PLUS extends 15 mm further but will not lock unless there is open space ahead. Fixed lengths: choose for body habitus. Know the hub release (older: quarter turn; newer: unscrew). Advance to the artery's medial border. Snug tunnel means less leakage. If the catheter meets resistance, the cannula tip may be kinked: withdraw slightly.

With any unfamiliar set, test-feed the catheter first and note where the tip emerges.

Pre-op or post-op?

Pre-op suits theatre flow. **Post-op in recovery** avoids the surgical field and intraoperative dislodgement, and allows a standard transverse approach (still enter at the lateral edge of sartorius, not through it). A padded hub under a tourniquet has not been a problem in Chin's practice.

4. Fixation: the most important step

1. Site **clean and dry**: be sparing with gel from the start.
2. **Tissue glue plug** at the exit site: seals the hole, stops backward leak, controls ooze. Let it dry.
3. **Epi-Guard**, foam under the hub, catheter running obliquely towards the hip with no twist or kink.
4. One or two **small Tegaderms**; keep the footprint small.
5. **Micropore window frame** round every Tegaderm edge, flat and wrinkle-free, then **wipe with alcohol**: it stays on even when wet.
6. Tape the remaining length. Fix filter and hub on the **lateral lower abdomen above the waistband**. Wrap the connector to prevent disconnection.

5. What to tell the patient

- Single shot lasts roughly 12 to 24 h; catheter with an elastomeric pump about **3 days**.
- Numbness: medial knee, down the medial calf to the ankle. Expect pain still from the **thigh, back of knee and lateral knee**, plus joint inflammation, so regular simple analgesia continues.
- **Falls**: walking aid, someone at home, no driving, brace if prescribed.
- No heat or ice directly on numb skin. Sponge baths only. Pump **at or above** catheter level; no kinks or clamps.
- **Leak**: common and not dangerous; a firm bandage round the thigh often stops it. **Disconnection**: never reconnect; the catheter comes out.
- **Report at once**: ringing ears, metallic taste, numb lips, seizure.
- **Removal at home** with the pain team on the phone: pull straight out, check the tip is intact, press for 3 minutes, plaster.

6. The rest of Station 4

General teaching points, not drawn from the videos. Follow your local guidelines.

Block	Key points
Femoral	Inguinal crease, above the profunda. Nerve lateral to artery, deep to fascia iliaca , on iliopsoas. Spread must be under fascia iliaca. Typically 10 to 20 mL. Quadriceps weakness: falls risk.
LFCN	Just medial and inferior to ASIS, in the fat-filled tunnel between sartorius and tensor fasciae latae. Sensory only; 3 to 5 mL. Anatomy varies.

Watch the source videos

Ki-Jinn Chin, "Adductor Canal Catheters" playlist: youtube.com/playlist?list=PLrTgRae0xIkOk9tXEZp_5wXNBAF9vkV7a (Arrow StimuCath, Pajunk SonoLong and E-Cath PLUS how-to guides, plus two patient guides.)

Regional Anesthesiology and Acute Pain Medicine, "Adductor Canal Block": youtube.com/watch?v=MfdaL95f7mc

ISAFE approach: Dos Santos Fernandes et al. J Clin Anesth 2022;76:110571.

Teaching summary prepared for ARACUS 2026 candidates. It paraphrases publicly available videos and does not replace supervised training or local guidelines. Drug choices and doses are for your own institution to decide.